

Italo Rojas

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PROFESSIONAL SUMMARY

Audio software engineer with an MS in Media Arts & Technology (UCSB), building real-time audio tools in C++/JUCE and working with neural audio synthesis. Currently exploring frameworks for building real-time conversational AI agents. Before graduate school, co-founded and led a consultancy for four years (50+ projects, 40+ clients, \$0 to ~\$120K USD annual revenue). Brings a product mindset, cross-functional leadership, and client communication to audio engineering teams.

CORE COMPETENCIES

- Frontend Development
 - Real-Time Systems Design
 - Cross-Functional Project Leadership
 - Data-Driven Decision Making
 - Client & Stakeholder Management
 - Rapid Prototyping
 - Consulting & Delivery Management
 - Applied Audio ML
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JOB EXPERIENCE

Co-founder & COO | Realiza — Sustainability Consulting, Chile *2021–2025*

- Co-founded and led a sustainability consultancy delivering 50+ projects for 40+ clients across Chile and internationally, including Universidad de Aysén, PUC Chile, and AJE Group's Bio Amayu supply chain, growing revenue from \$0 to ~\$120K USD annually.
- Designed data pipelines aligned with GHG Protocol and SBTi, applying AHP and Monte Carlo simulation for project prioritization and emissions forecasting.
- Defined the product and data model for VestaESG, a carbon-accounting SaaS platform (React, FastAPI, PostgreSQL), coordinating developers while building the frontend.

Project Manager | UC Sustainability Directorate, Chile *2018–2020*

- Led the design, data analysis, and implementation of sustainability initiatives in transportation, biodiversity, and waste management at the top-ranked university in Latin America (QS Latin America University Rankings 2026).
- Managed large-scale environmental datasets to produce a Carbon Neutrality Strategy roadmap — one of the first in Chilean higher education.

Data Analyst | GlocalMinds, Chile *2021*

- Mapped and analyzed organizations across Ibero-America associated with new economies, transforming complex network data into accessible visual formats to highlight systemic impact.

SKILLS

Technical Skills: Real-time audio processing, digital signal processing, neural audio synthesis, gesture/object-detection interaction design, spatial audio design, data pipeline design

Tools & Automation: C++, JUCE, Max/MSP, SuperCollider, Python, JavaScript/TypeScript, React, Svelte, SQL, FastAPI, PostgreSQL, Docker, Git

Languages: Spanish (Native), English (Advanced – C1, IELTS)

EDUCATION

MS, Media Arts and Technology | *University of California, Santa Bárbara 2026*

Focus: DSP audio, interactive media, generative systems, science communication through art.

Industrial Civil Engineer | *Pontificia Universidad Católica de Chile 2013–2019*

Major in Transportation Systems - Externalities & Environmental Impact. Minor in Water and Energy.

MEDIA ART PROJECTS

AquiFuturo (UCSB Thesis) — Python, Max/MSP, RAVE, JUCE

AR mobile app (Unity, C#, AR Foundation) that visualizes a tree's hidden root system in augmented reality. Built the full pipeline: Python CLI tools for data processing, schema-validated spatial graph, and a real-time neural audio engine (RAVE) driven by root topology.

PostTalk — MediaPipe, C++, JUCE, Svelte, TypeScript

Gesture-driven audio effects controller for live performance: hand posture, tracked via MediaPipe, controls a 56-parameter reverb engine (FDN, early reflections, diffusion network) over a lock-free UI/DSP thread bridge.

The Dance of Laplace — C++, Allolib, GLSL, Ableton

Gray-Scott reaction-diffusion simulation with a Bi-Laplacian extension, rendered on a sphere and spatialized across the AlloSphere's speaker array for UCSB's End of Year Show 2026; audio stems routed to independent spatialization buses.

ADDITIONAL

Teaching: Lecturer, "Sustainability" — Institute for Sustainable Development, PUC Chile (2021),

Volunteering: Cverde NGO — Partner, Director of Finance (2024) · Mapu Kufüll Mushroom Cultivation Cooperative - Partner, Workshop Facilitator (2022–2024) · Kingdom Fungi Festival - Operations Team & Art Direction (2024)

Award: UC Young Leader — Sustainability Area, PUC Chile (2019)